What we're doing

Scoring the three checkpoints issue #169 selected — the final and early-stop winners of #117 (1.5B, 16 epochs) and the #146 3B — on the 554-protein contact eval set, and reporting **R-precision** for each.

The measurement spec is unchanged from what the project already publishes: exp89's ground truth, candidate-pair universe and metric implementation; exp82's rollout+resample inference recipe (100 resampled contacts-v1 rollouts per protein, top-k off), run through exp82's worker unmodified.

Why

All three checkpoints were picked by `eval/tokenized/contacts-v1-val/loss`, and they sit within **0.008 nats** of each other. So the question underneath "what R-precision do they get?" is whether val loss is still a useful selection signal at that granularity, or whether the loss to accuracy relationship has gone flat.

The prior evidence is a steep slope: 2.7566 to 2.7037 (0.053 nats) bought +0.11 R-precision. Extrapolated locally that predicts +0.016 for the 0.0076-nat gap between the two #117 checkpoints — small, but resolvable with a paired test over 554 proteins.

How the comparison is kept honest

Every checkpoint scores the same 554 proteins, so differences are reported **paired**. That matters: the between-protein spread of R-precision is ~0.3, so the unpaired SEM (~0.013) is wider than the entire effect being measured. The paired standard error is an order of magnitude smaller, because protein-to-protein variance cancels.

Two other guards. The #117 final checkpoint is re-scored rather than reused from its published run, so all three numbers come from one submission — and reproducing the published 0.535 is itself the harness check. And `verify_prepared_exports.py` gates the run on the three checkpoints sharing one vocabulary and one set of special-token ids, since a silent tokenizer shift would not fail loudly; it would just produce confident wrong numbers.

Results

554/554 proteins for all three checkpoints, 0 of 166,200 rollouts truncated.

checkpoint	val loss	R (all)	R (long)	AUC (all)
#117 · 1.5B · final	2.7037	**0.534**	**0.482**	0.933
#117 · 1.5B · early stop	2.6961	0.532	0.481	0.933
#146 · 3B · E8	2.7025	0.512	0.459	0.925

The #117 final row reproduces its published 0.5350 to 0.0006 — the harness check.

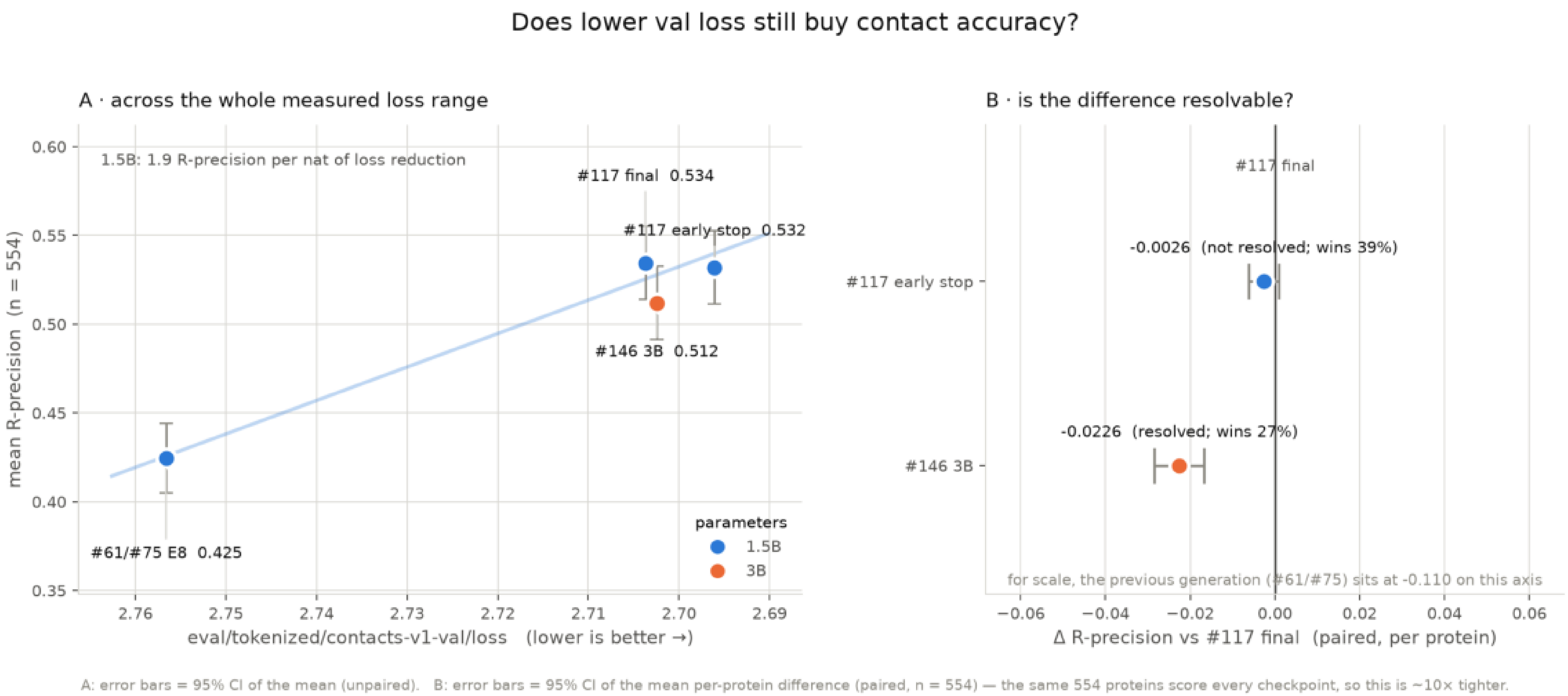
What it says

Early stopping on val loss bought nothing. The early-stop checkpoint has 0.0076 lower loss and is indistinguishable on contacts: Δ +0.0026 favouring the final checkpoint, CI [-0.0010, +0.0062], win rate 48.6%. The exp82 slope predicted +0.016. The loss to accuracy relationship is steep across training generations and flat inside one run's last 2,000 steps.

Matched val loss does not mean matched contact accuracy across model sizes. The 3B's loss is 0.0012 *better* than the 1.5B final, and its R-precision is 0.023 *worse* — resolvable, CI [+0.017, +0.028], and the largest effect in the comparison. (Confounded with epochs 8 vs 16 and wd 0.4 vs 0.2, so it is about this checkpoint, not scale as such.)

Practical consequence: `contacts-v1-val/loss` is not a usable tie-breaker below ~0.01 nats, and must not be compared across model sizes to pick a contact predictor.

loss_vs_rprecision.png



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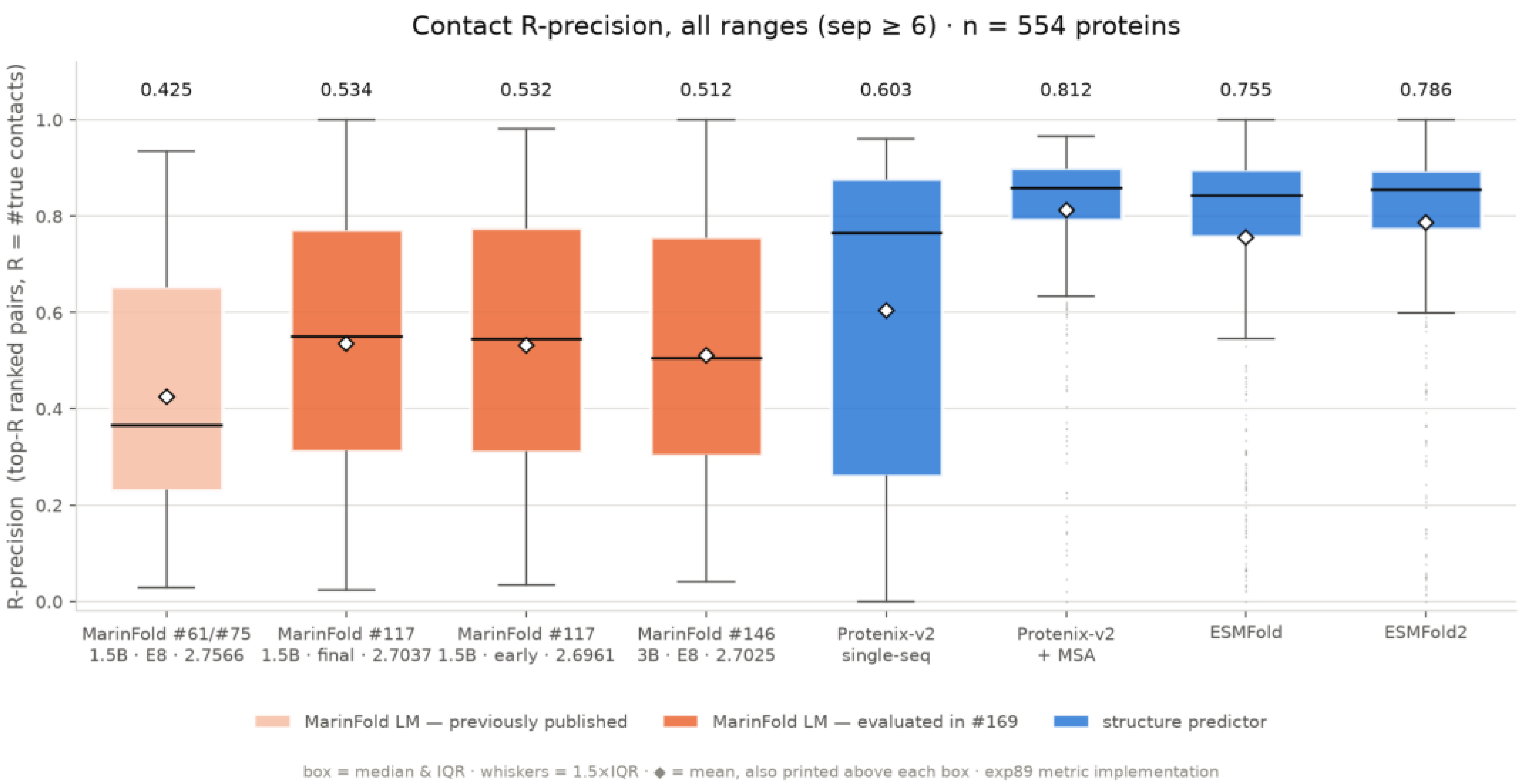
where_we_stand_rprecision.png



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